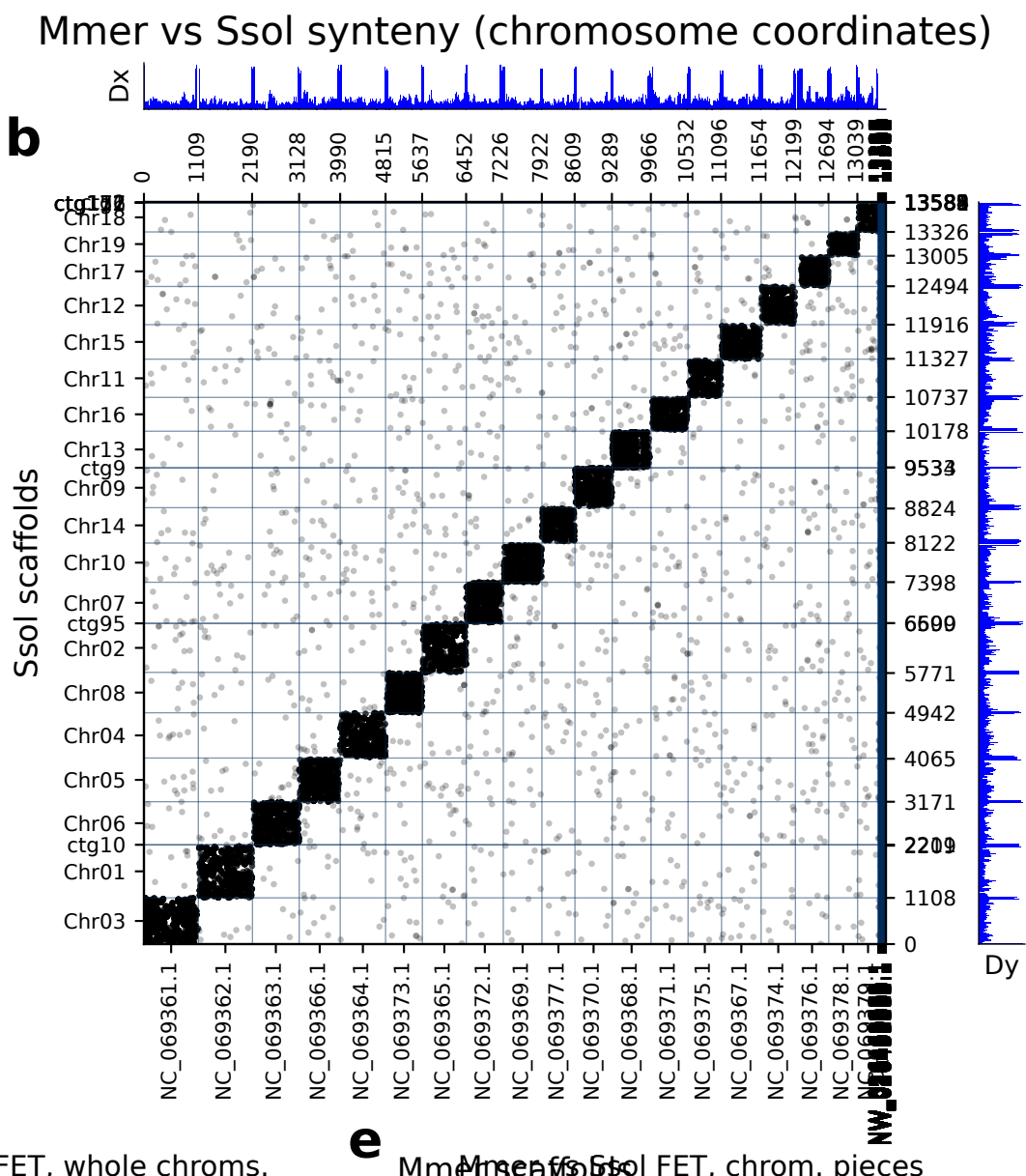
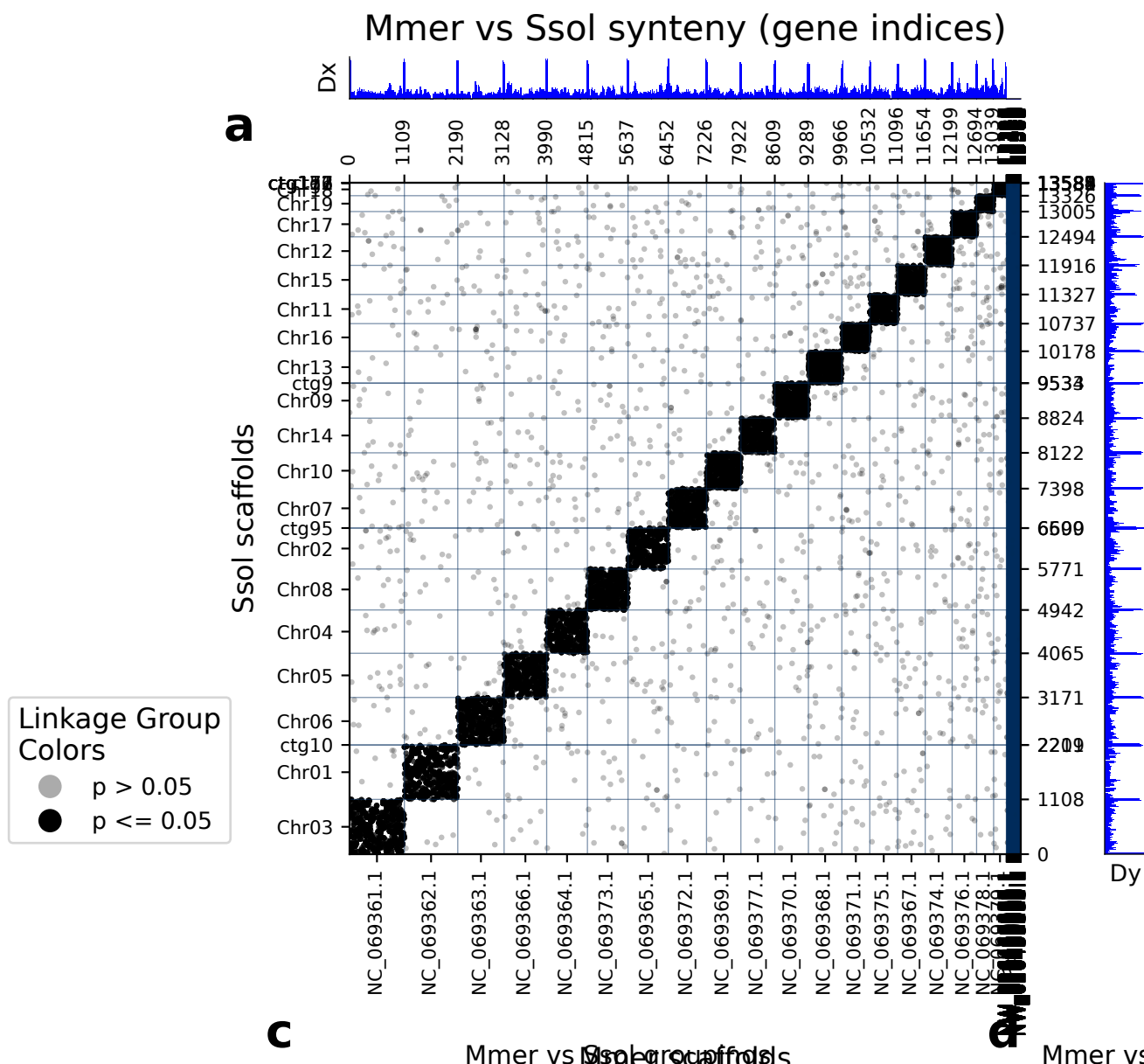


diamond: 13590 orthologs on 282 Mmer scaffolds and 27 Ssol scaffolds.



C Mmer vs Mneg scaffolds

Mmer vs Ssol FET, whole chroms.

	Mmer scaf	Ssol scaf	p (FET)	Count
0	NC_069362.1	Chr01	0.00e+00	997
1	NC_069361.1	Chr03	0.00e+00	995
2	NC_069363.1	Chr06	0.00e+00	856
3	NC_069366.1	Chr05	0.00e+00	767
4	NC_069364.1	Chr04	0.00e+00	741
5	NC_069373.1	Chr08	0.00e+00	734
6	NC_069365.1	Chr02	0.00e+00	713
7	NC_069372.1	Chr07	0.00e+00	691
8	NC_069377.1	Chr14	0.00e+00	619
9	NC_069369.1	Chr10	0.00e+00	615
10	NC_069370.1	Chr09	0.00e+00	605
11	NC_069368.1	Chr13	0.00e+00	594
12	NC_069375.1	Chr11	0.00e+00	497
13	NC_069374.1	Chr12	0.00e+00	482
14	NC_069367.1	Chr15	0.00e+00	475
15	NC_069371.1	Chr16	0.00e+00	474
16	NC_069376.1	Chr17	0.00e+00	404
17	NC_069378.1	Chr19	0.00e+00	277
18	NC_069379.1	Chr18	0.00e+00	201
19	NC_048487.1	ctg177	6.37e-07	3
20	NC_048487.1	ctg116	1.73e-03	2

e Mmmer ffs Sol FET, chrom. pieces

	Mmer scaf	Ssol scaf	p (FET)	Count
0	NC_069362.1 : 0-end	Chr01 : 0-end	0.00e+00	997
1	NC_069361.1 : 0-end	Chr03 : 0-end	0.00e+00	995
2	NC_069363.1 : 0-end	Chr06 : 0-end	0.00e+00	856
3	NC_069366.1 : 0-end	Chr05 : 0-end	0.00e+00	767
4	NC_069364.1 : 0-end	Chr04 : 0-end	0.00e+00	741
5	NC_069373.1 : 0-end	Chr08 : 0-end	0.00e+00	734
6	NC_069365.1 : 0-end	Chr02 : 0-end	0.00e+00	713
7	NC_069372.1 : 0-end	Chr07 : 0-end	0.00e+00	691
8	NC_069377.1 : 0-end	Chr14 : 0-end	0.00e+00	619
9	NC_069369.1 : 0-end	Chr10 : 0-end	0.00e+00	615
10	NC_069370.1 : 0-end	Chr09 : 0-end	0.00e+00	605
11	NC_069368.1 : 0-end	Chr13 : 0-end	0.00e+00	594
12	NC_069375.1 : 0-end	Chr11 : 0-end	0.00e+00	497
13	NC_069374.1 : 0-end	Chr12 : 0-end	0.00e+00	482
14	NC_069367.1 : 0-end	Chr15 : 0-end	0.00e+00	475
15	NC_069371.1 : 0-end	Chr16 : 0-end	0.00e+00	474
16	NC_069376.1 : 0-end	Chr17 : 0-end	0.00e+00	404
17	NC_069378.1 : 0-end	Chr19 : 0-end	0.00e+00	277
18	NC_069379.1 : 0-end	Chr18 : 0-end	0.00e+00	201
19	NC_048487.1 : 0-end	ctg177 : 0-end	6.37e-07	3
20	NC_048487.1 : 0-end	ctg116 : 0-end	1.73e-03	2

(a) depicts the chromosomes/scaffolds of Mmer (x-axis) plotted against the chromosomes/scaffolds of Ssol (y_axis). Each dot in the plot represents an ortholog, specifically a reciprocal best diamond blastp match between two species. The unit of the x- and y-axes are the number of orthologous proteins between these two species: 13590 orthologs found between 282 Mmer scaffolds and 27 Ssol scaffolds. If there are chromosome breaks, Fisher's exact test is used to calculate the significance of the interactions between the sub-chromosomal pieces. Otherwise Fisher's exact test is calculated on whole chromosomes. The opacity of the dots depict the significance from Fisher's exact test. Dots that are a solid color are in cells with a FET p-value less than or equal to 0.05. Dots that are translucent are in cells with a FET p-value greater than 0.05. The Dx and Dy values depict places where there may be sudden breaks in synteny. See the supplementary information the following paper for more information on Dx and Dy: Simakov, Oleg, et al. Deeply conserved synteny resolves early events in vertebrate evolution. *Nature Ecology & Evolution* 4.6 (2020): 820-830. (b) depicts the same information as panel a, but it is plotted in the organisms chromosome basepair coordinates rather than gene index. This is useful for visualizing gene-poor regions of the chromosomes (c) shows which gene groups are most prevalent in each chromosome pair. The FET p-value in this table corresponds to the whole-chromosome FET p-value, and is not a FET value of the correlation between the chromosome pair and the gene group. (d) shows chromosome-scale significance values. This table shows the same information as c, but does not factor in gene group information. (e) shows the FET p-values of the sub-chromosomal compartments. Useful for showing if single arms of chromosomes are correlated with other regions.